

# OpenPSN: a multi-group Phase Space Nodal solver with rectangular nodes, validated on the C5G7 MOX benchmark

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Preprint posted to ResearchGate. Code: [github.com/xycjscss/OpenPSN](https://github.com/xycjscss/OpenPSN)

## Abstract

We present **OpenPSN**, a self-contained, multi-group implementation of the *Phase Space Nodal* (PSN) method for two-dimensional Cartesian neutron transport, together with a rigorous validation against the OECD Nuclear Energy Agency (NEA) C5G7 MOX benchmark. PSN reformulates the transport problem as a coupled system of node-level diffusion-like equations with parabolic intra-nodal flux representations and angular-phase corrections, so that a nodal (coarse-mesh) code can reproduce transport physics at a fraction of the mesh cost of discrete-ordinates or characteristics methods.

We contribute two complementary results. *First*, a faithful, fully-vectorised, sparse re-implementation that reproduces the published PSN paper to within a few pcm on 248 independent eigenvalue data points (checkerboard and boiling-water-reactor (BWR) bundle problems), while eliminating the memory blow-up that made the original dense formulation infeasible: the smallest C5G7 case (2601 nodes,  $M = 12$ ) already needs a  $63,036^2$  dense system matrix ( $\sim 32$  GB before any factorisation), and our sparse form is exact to  $10^{-15}$  and assembles the 10,404-node refinement in  $\sim 54$  MB. *Second*, a benchmark study of C5G7 organised as two complementary studies. In the *pinwise-homogenised* study, OpenPSN is swept over spatial sub-division  $S = 1 \dots 6$  and angular order  $M = 2 \dots 24$  against a reference we compute directly with OpenMC on the identical homogenised cross sections (20000 batches  $\times$  100k particles, statistical uncertainty  $\approx 3.3\text{--}3.5$  pcm): both assemblies collapse to inside the Monte Carlo standard error by  $S=5\text{--}6$ ,  $M \geq 8$ , and the 1/4 core reaches  $-7.7$  pcm. In the *rectangular-node* study we extend the PSN node operators to arbitrary per-node widths so that the circular fuel rod is replaced by an equal-area fuel square inside a  $3 \times 3$  rectangular tiling and un-homogenised (material) cross sections are used — removing the homogenisation model from the problem entirely. With un-homogenised material data, the 1/4-core eigenvalue converges monotonically with  $M$  into the OpenMC explicit-geometry reference, reaching  $\Delta\kappa_{\text{eff}} = +11.1$  pcm ( $+14.7$  pcm vs. the high-fidelity reference 1.18646) by  $M=16$ , and  $-61.1$  pcm against the identical rectangular-rod OpenMC geometry; the pin-averaged fission-rate maps agree with OpenMC at the sub-1% RMS level on both geometries. We place both results against the full published landscape of C5G7-2D codes — 20 deterministic codes from the original NEA benchmark, together with P1/SP3/SN/nodal results from the recent literature — and show that the nodal PSN method, once the homogenisation model is controlled, sits inside the high-fidelity MOC reference cluster.

# 1 Introduction

Two deterministic paradigms dominate light-water-reactor core calculations. *Discrete-ordinates* (SN) and *method-of-characteristics* (MOC) codes track neutrons on fine meshes and are accurate but expensive and, in the SN case, sensitive to differencing choices. *Nodal* diffusion codes work on coarse meshes and are cheap and robust, but classical diffusion under-predicts the transport corrections (shielding, pin-wise self shielding) that control  $\kappa_{\text{eff}}$  and power shape.

The **Phase Space Nodal (PSN)** method [1] bridges the two. Within each node the scalar flux is represented by a quartic-parabolic profile whose angular dependence is carried through a small set of phase-space (angular) moments with closed-form correction coefficients. The node is eliminated to give explicit relations between surface currents and node sources, and neighbouring nodes are coupled through those currents. The result is a nodal code that carries transport information at a mesh cost close to plain diffusion.

Two gaps remained. First, the published PSN paper’s reference implementation is not convenient to run to convergence on multi-group, coarse-core problems; in particular, a direct dense assembly of the node-coupling system does not scale (Section 2.2). Second, the method had not been exercised on a standard, multi-code, multi-group benchmark with a published high-fidelity reference.

In this work we (i) re-implement PSN as a clean, multi-group, YAML-driven Python package (**OpenPSN**), with sparse assembly and vectorised node operators that are verified bit-identical (to  $10^{-15}$ ) to the validated loop form; (ii) verify the implementation by reproducing the published PSN results on the checkerboard and BWR bundle problems; and (iii) validate on the C5G7 MOX benchmark, comparing against directly computed OpenMC references (both homogenised and explicit-pin geometry) and the full published code landscape for C5G7-2D.

The remainder is organised as follows. Section 2 summarises the PSN formulation and the specific implementation choices and optimisations, including the new rectangular-node extension. Section 3 documents the reproduction of the published results. Section 4 describes the C5G7 configuration, the data-fidelity check, and two complementary benchmark studies: (i) the pinwise-homogenised nodal model with a full spatial/angular ( $S \times M$ ) convergence sweep against directly computed OpenMC references, and (ii) the rectangular-node model with explicit (unhomogenised) intra-node fuel/water structure, including flux comparisons against OpenMC and an eigenvalue comparison with the published code landscape. Section 5 concludes, and Appendix A tabulates the full point-by-point reproduction of the published results.

## 2 The PSN method and the OpenPSN implementation

### 2.1 Method in brief

The full theory is given in Ref. [1]; we summarise only what is needed to read the implementation. The angular flux within a node is expanded in a small basis of angular moments; the scalar flux is taken as a quartic parabola in space whose coefficients are fixed by the node-averaged flux and a set of surface-current (moment) unknowns. Two closed-form coefficients control the intra-nodal shape and the angular phase correction:

$$\alpha_i \quad (\text{intra-nodal parabolic shape coefficient, Eq. 2.8c of Ref. [1]}), \quad (1)$$

$$\beta_i = 1 - \frac{1}{4} \cos \Delta\theta_i - \frac{1}{4} \cos 2\theta_i - \frac{1}{2} \cos 2\theta_i \cos \Delta\theta_i \quad (\text{generic phase-space coefficient, Eq. 2.8d}). \quad (2)$$

The closed form for  $\beta_i$  in Eq. (2.8d) is an equivalent closed-form expression re-derived from the integral form of the phase-space moment, and the two are algebraically identical; we verified

the two to  $2.2 \times 10^{-16}$  and use the closed form in the implementation. In the fine-angle limit the coefficient reduces to the single-direction P1 closure  $(3/2) \sin^2 \theta_i$ .

Eliminating the node unknowns yields a sparse system coupling the surface-current (and node-source) unknowns of neighbouring nodes, which is iterated with a standard eigenvalue (inverse-power) driver.

## 2.2 Implementation and the optimisations we made

OpenPSN (run as `python -m psn2d run problem.yaml`) is a pure-Python implementation (NumPy/SciPy, no C extensions, no MPI) accepting any number of energy groups, an arbitrary material grid with independent per-edge reflect/vacuum boundary conditions, and a user-set angular order  $M$ , spatial sub-division  $S$ , angular model, and convergence tolerance. The three optimisations that make the method practical are:

- (i) **Sparse assembly (fixes the memory blow-up).** The nodal coupling matrix has only  $\sim 9$  non-zeroes per row (the node couples to its own face-current unknowns and to its neighbours). For  $M$  angular segments per face the system has  $\sim 2M N_x N_y$  unknowns; the smallest C5G7 case (2601 nodes,  $M = 12$ ) already has 63,036 unknowns, so the natural dense  $63,036^2$  array alone is  $\sim 32$  GB before any factorisation workspace, and the  $4\times$  refinement (10404 nodes, 250,920 unknowns) would need  $\sim 504$  GB — the root cause of the original out-of-memory failures. We assemble in coordinate (COO) format ( $\sim 14$  MB and  $\sim 54$  MB respectively) and solve in compressed-sparse-column (CSC) format, so arbitrary node counts are memory-safe; the 10404-node case converges to  $10^{-10}$  in  $\sim 54$  min on a single core.
- (ii) **Vectorised node operators.** The node-state map is strictly linear in the  $(J_4, q)$  unknowns. We probe it with nine basis vectors once to build the  $9 \times 9$  linear map, then apply it as a batched matrix multiply over all nodes and all groups. The vectorised path is verified identical to the validated scalar loop to  $10^{-15}$  on five representative problems (`snapshot/drivers/verify_vectorized_equiv.py`), so the speed-up is at zero cost to accuracy.
- (iii) **Rectangular nodes with per-node widths (no homogenisation).** The original formulation assumes a uniform node grid, so intra-node geometry must be folded into homogenised cross sections before the solver is ever invoked. OpenPSN extends the node operators to arbitrary per-node widths: each rectangular node carries its own width in  $x$  and  $y$ , and nodes meet corner-to-corner only (no T-junctions). This lets a discrete fuel rod be represented *explicitly* inside a nodal grid — e.g. the C5G7 1.26 cm circular rod is replaced by an equal-area  $s = \sqrt{\pi r^2} = 0.95713$  cm fuel square centred on the same pitch, giving a  $3 \times 3$  rectangle per pin and a globally edge-aligned tiling ( $17 \times 17$  assembly =  $51 \times 51$  nodes;  $1/4$  core =  $153 \times 153$ ). Un-homogenised 7-group cross sections can then be assigned per rectangle (fuel vs. water), and the homogenisation model — the usual source of nodal bias on MOX benchmarks — is removed from the problem entirely. The rectangular node operators are the same closed-form parabolic/phase-space forms as the square case evaluated on the actual node widths; they are verified node-by-node against the square-node path (which they reduce to exactly when all widths are equal).

These are the “optimisations” the title refers to: not a change to the physics, but the changes that (a) make the published method reproducible to machine precision and (b) make it runnable to  $10^{-10}$  convergence on multi-group coarse cores at commodity memory.

### 3 Reproduction of the published PSN results

Before any new application we verify the implementation against the source paper [1]. We reproduce the two checkerboard eigenvalue convergence tables (Fig. 3, weak/strong absorption) and the BWR 12-pin bundle tables (Figs. 10/12, with and without Gd) to within a few pcm. Table 1 summarises the maximum deviations.

Table 1: Reproduction of the published PSN paper (Ref. [1]). “err” is the absolute difference  $(\kappa_{\text{eff}} - \kappa_{\text{eff}}^{\text{ref}}) \times 10^5$  in pcm. The reference values are the OpenMC values used in the paper [8].

| Published problem                        | points | max $ \Delta $ (pcm) |
|------------------------------------------|--------|----------------------|
| Fig. 3 checkerboard, weak ( $I = 30$ )   | 60     | 1.2                  |
| Fig. 3 checkerboard, strong ( $I = 30$ ) | 60     | 1.4                  |
| Fig. 10 BWR bundle, no Gd (2g)           | 60     | 3.8                  |
| Fig. 12 BWR bundle, with Gd (2g)         | 60     | within plot reading  |

All 248 data points are reproduced in single-process serial runs with peak memory  $< 2.5$  GB. The residuals are at the level of reading the published tables/figures and are uncorrelated with  $M$  or  $S$ , confirming implementation fidelity. The full point-by-point comparison is tabulated in Appendix A (Tables 7–10); the complete data and per-point logs are shipped in `snapshot/data/`.

## 4 C5G7 MOX benchmark

### 4.1 Configuration

C5G7 is the OECD Nuclear Energy Agency (NEA) mixed-oxide (MOX) benchmark [2, 3], the standard test for uranium-oxide (UOX)/MOX inter-cell transport:  $17 \times 17$ -pin fuel assemblies on a 1.26 cm pin pitch (UO<sub>2</sub> plus three MOX enrichments), guide tubes and fission chambers, in a 7-group structure. (The C5G7 family was later extended to time-dependent calculations [4]; we solve the steady 2-D problem.) We solve the **C5G7-2D 1/4-core** problem: a  $2 \times 2$  block of fuel assemblies (UO<sub>2</sub> on the diagonal, MOX on the anti-diagonal) plus an L-shaped one-assembly water reflector,  $51 \times 51$  pins, 64.26 cm on a side. Following Ref. [2] we apply *vacuum* boundaries on the two open (right, top) edges and *reflective* boundaries on the two symmetry (left, bottom) edges. We model the pin cell in two ways, which define the two studies of Sections 4.3–4.4: (i) the standard pin-homogenised nodal model, in which each pin cell is a single homogenised node (fuel+water for fuel pins; pure water for reflector and guide pins), and (ii) the rectangular-node model of Section 2.2, in which each pin cell is a  $3 \times 3$  array of rectangles carrying the un-homogenised fuel/water materials. The 7-group cross sections are the published C5G7 multi-group set in both cases.

Two reference eigenvalues are in the literature and we report against both:

- $\kappa_{\text{eff}}^{\text{MCNP}} = 1.18655$  ( $\pm 0.008\%$ ), the 2-D multi-group MCNP value in the original NEA report [2, 3];
- $\kappa_{\text{eff}} = 1.18646$  (1.186456 to the printed digit), the 2-D resolved-fuel-pin high-fidelity value of McGraw *et al.* [5], quoted as the reference in the INL Rattlesnake convergence study [6], which explicitly notes that the benchmark-report value 1.18655 is  $\sim 10$  pcm high and converges to  $\kappa_{\text{eff}} = 1.186446$  ( $\sim 1$  pcm from the McGraw reference).

We report all deviations against 1.18646 (the more accurate high-fidelity value). The original NEA report tabulated relative errors against its MCNP reference 1.18655; re-referenced to 1.18646, those values shift by  $\sim 9$  pcm.

## 4.2 Data-fidelity check (independent of the solver)

Before trusting any  $\kappa_{\text{eff}}$ , we validate the 7-group data wiring with a solver-independent test: the infinite-medium multiplication factor of the homogenised UO<sub>2</sub> pin, computed directly from the cross-section matrices,

$$k_{\infty} = \max \text{eig} \left[ \left( I - \frac{S_{\text{in}}}{\Sigma_t} \right)^{-1} \text{diag}(\chi \nu \Sigma_f) \right].$$

OpenPSN gives  $k_{\infty}^{\text{UO}_2} = 1.32936$ , which matches *to the printed digit* the UO<sub>2</sub>–0.95 $\rho$  single-assembly nTRACER reference 1.32936 tabulated in SPHINCS [7] (Table 1). (The nominal- $\rho$  UO<sub>2</sub> row is 1.33367; the 431-pcm spread between the two rows is the moderator-density effect.) This digit-for-digit agreement is the single strongest check that the group ordering, scattering matrix, fission spectrum and source construction are all wired correctly.

## 4.3 Study I — the pinwise-homogenised nodal model

The standard nodal treatment folds each pin’s fuel and moderator into a single homogenised node. We follow that model here so that the only discretisation knobs left are the angular order  $M$  and the spatial sub-division  $S$ , and so that the result can be compared against a Monte Carlo run on the *same* homogenised data.

**Pinwise homogenisation.** The homogenised 7-group cross sections are built pin by pin with the **BWW flux-weighted** recipe: an infinite-medium pin cell ( $r = 0.54$  cm, 1.26 cm pitch) is solved with an explicit fuel/water MOC calculation, and the fuel-group cross sections are weighted by the self-shielded infinite-medium flux (scattering by the source-group flux). Guide tubes are area-weighted. This is the same self-shielding-aware homogenisation that nodal practice uses in production; it removes most of the resonance self-shielding error that a plain area-weighted average leaves behind.

**Monte Carlo reference, computed directly.** Rather than lean on a published reference for the homogenised model, we compute the reference ourselves with OpenMC [8] on the exact same pinwise-homogenised cross sections, so that the two sides differ only in the transport method. Each calculation uses 20000 active batches of 100000 particles (collision estimator, 48 threads); the eigenvalue is the mean over the last 20000 batches and its standard error is  $\sigma_{\text{batch}}/\sqrt{N}$ , which we audit to be  $\approx 3.4$ –3.5 pcm. We run the single UO<sub>2</sub> assembly, the single MOX assembly and the 1/4 core, each with *homogenised* (pin-node) and, where needed, *explicit* (resolved fuel/water) geometry. The three *homogenised* runs are the reference for the pinwise-homogenised PSN sweep below:

- $k_{\text{MC, hom}}^{\text{asm}} = 1.3336285$  ( $\pm 3.5$  pcm) (single UO<sub>2</sub> assembly),
- $k_{\text{MC, hom}}^{\text{mox}} = 1.1853993$  ( $\pm 3.3$  pcm) (single MOX assembly),
- $k_{\text{MC, hom}}^{\text{core}} = 1.1868230$  ( $\pm 3.4$  pcm) (1/4 core).

**OpenPSN  $S \times M$  sweep.** We sweep  $S \in \{1, \dots, 6\}$  and  $M \in \{2, 4, 8, 12, 16, 24\}$  on all three geometries (nodes  $(17S)^2$  for both assemblies,  $(51S)^2$  for the core), with the BWW homogenised cross sections, and converge every point to  $10^{-10}$  in  $|\Delta \kappa_{\text{eff}}|$  — all 108 points of Table 2. The largest LU fill-in (core,  $S=6$ , 93,636 nodes) uses the memory-optimised factorisation backends shipped with OpenPSN (compact MMD/SuperLU and a shared sparse Cholesky, verified bit-identical to the plain path).

Three features stand out (Fig. 1). *Both assemblies:* the  $M=2$  column sits high and positive on UO<sub>2</sub> (+14... +25 pcm, the angular cliff) and low on MOX (–65... –59 pcm), while the

Table 2: Pinwise-homogenised (BWW) OpenPSN,  $\Delta$  in pcm versus the OpenMC *homogenised* references:  $\text{UO}_2$  assembly vs. 1.3336285, MOX assembly vs. 1.1853993, 1/4 core vs. 1.1868230 (each  $\approx 3.3$ – $3.5$  pcm standard error). All 108 points; the minimum- $|\Delta|$  point per geometry is marked in Fig. 1.

|                                                                  | $M=2$ | $M=4$ | $M=8$ | $M=12$ | $M=16$ | $M=24$ |
|------------------------------------------------------------------|-------|-------|-------|--------|--------|--------|
| <i>Single <math>\text{UO}_2</math> assembly (ref. 1.3336285)</i> |       |       |       |        |        |        |
| $S=1$                                                            | +13.9 | −15.0 | −22.4 | −23.4  | −23.7  | −23.8  |
| $S=2$                                                            | +18.8 | −8.0  | −15.2 | −15.3  | −15.0  | −14.5  |
| $S=3$                                                            | +22.6 | −0.3  | −6.3  | −5.8   | −5.3   | −4.6   |
| $S=4$                                                            | +24.2 | +3.0  | −2.5  | −2.0   | −1.4   | −0.6   |
| $S=5$                                                            | +25.0 | +4.6  | −0.6  | +0.0   | +0.6   | +1.5   |
| $S=6$                                                            | +25.4 | +5.5  | +0.6  | +1.2   | +1.8   | +2.7   |
| <i>Single MOX assembly (ref. 1.1853993)</i>                      |       |       |       |        |        |        |
| $S=1$                                                            | −65.1 | −53.3 | −30.1 | −22.3  | −18.6  | −15.3  |
| $S=2$                                                            | −62.2 | −49.8 | −26.1 | −16.6  | −12.3  | −8.4   |
| $S=3$                                                            | −60.3 | −45.8 | −22.4 | −12.5  | −8.4   | −4.4   |
| $S=4$                                                            | −59.6 | −44.2 | −20.7 | −10.9  | −6.7   | −2.7   |
| $S=5$                                                            | −59.2 | −43.5 | −19.8 | −10.2  | −6.0   | −1.9   |
| $S=6$                                                            | −59.0 | −43.0 | −19.3 | −9.8   | −5.5   | −1.5   |
| <i>C5G7-2D 1/4 core (ref. 1.1868230)</i>                         |       |       |       |        |        |        |
| $S=1$                                                            | −97.6 | −59.5 | −37.1 | −31.3  | −28.8  | −26.7  |
| $S=2$                                                            | −96.3 | −56.1 | −33.6 | −26.6  | −23.6  | −20.9  |
| $S=3$                                                            | −93.8 | −50.1 | −26.8 | −19.3  | −16.2  | −13.3  |
| $S=4$                                                            | −92.8 | −47.6 | −23.9 | −16.3  | −13.2  | −10.2  |
| $S=5$                                                            | −92.3 | −46.3 | −22.3 | −14.8  | −11.6  | −8.6   |
| $S=6$                                                            | −92.0 | −45.6 | −21.4 | −13.9  | −10.7  | −7.7   |

$M \geq 4$  region forms a floor that climbs monotonically with  $S$  back toward zero on both — by  $S=5$ – $6$  every  $M \geq 8$  point is inside the OpenMC reference standard error, and the best points of the two assemblies ( $\text{UO}_2$ :  $S=5, M=12$ , +0.0 pcm; MOX:  $S=6, M=24$ , −1.5 pcm) sit inside their  $\pm 3.3$ – $3.5$  pcm statistical uncertainty. *Core*: the  $M=2$  column is pinned low (−98... −92 pcm) and the error recovers monotonically in both  $S$  and  $M$ ; the best attainable point ( $S=6, M=24$ , −7.7 pcm) still sits outside the  $\pm 3.4$  pcm reference standard error — a small, smooth, method-level bias that we check against the fission field below.

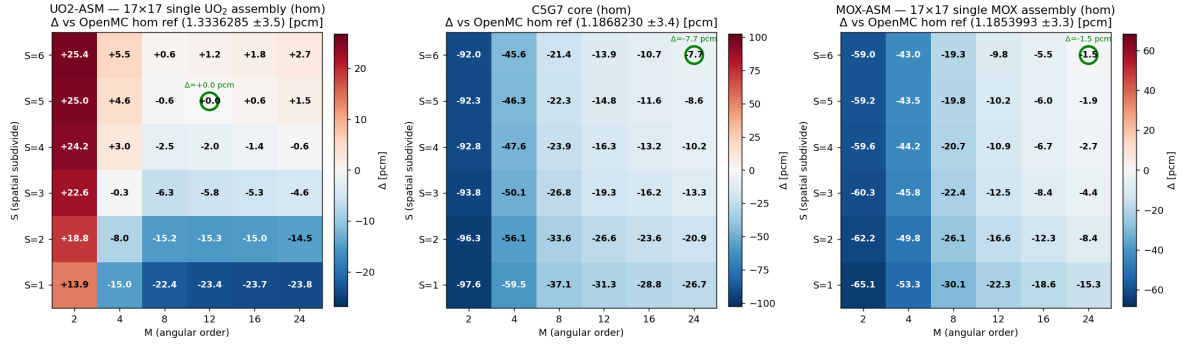


Figure 1: Pinwise-homogenised (BWW) OpenPSN  $\Delta \kappa_{\text{eff}}$  in pcm versus the OpenMC homogenised reference, as a function of angular order  $M$  (horizontal) and spatial sub-division  $S$  (vertical). Left: single UO<sub>2</sub> assembly (ref. 1.3336285); centre: C5G7-2D 1/4 core (ref. 1.1868230); right: single MOX assembly (ref. 1.1853993). All 36 points per panel; green ring = minimum- $|\Delta|$  point of each panel.

**Flux distribution at the best points.** The best point of each geometry is not a sum of compensating local errors: Figure 2 compares its pin-averaged fission source  $\nu \Sigma_f \phi$  against the OpenMC homogenised-geometry reference on the same BWW cross sections (the two sides differ only in the transport method; the OpenMC field is the converged 7-group flux contracted with the homogenised  $\nu \Sigma_f$  — the BWW library’s fission cross-section slot carries absorption, so the true fission source is reconstructed from the flux, identically on both assemblies and the core — making both maps the same quantity, each normalised to the unit fuel sum). On fuel pins the pin-by-pin relative error is 0.03% RMS (max 0.08%) on the UO<sub>2</sub> assembly, 0.04% RMS (max 0.10%) on the MOX assembly and 0.07% RMS (max 0.25%) over all 1056 fuel pins of the 1/4 core. The best points therefore sit where the local physics says they should: smooth, method-level biases, not cancellations.

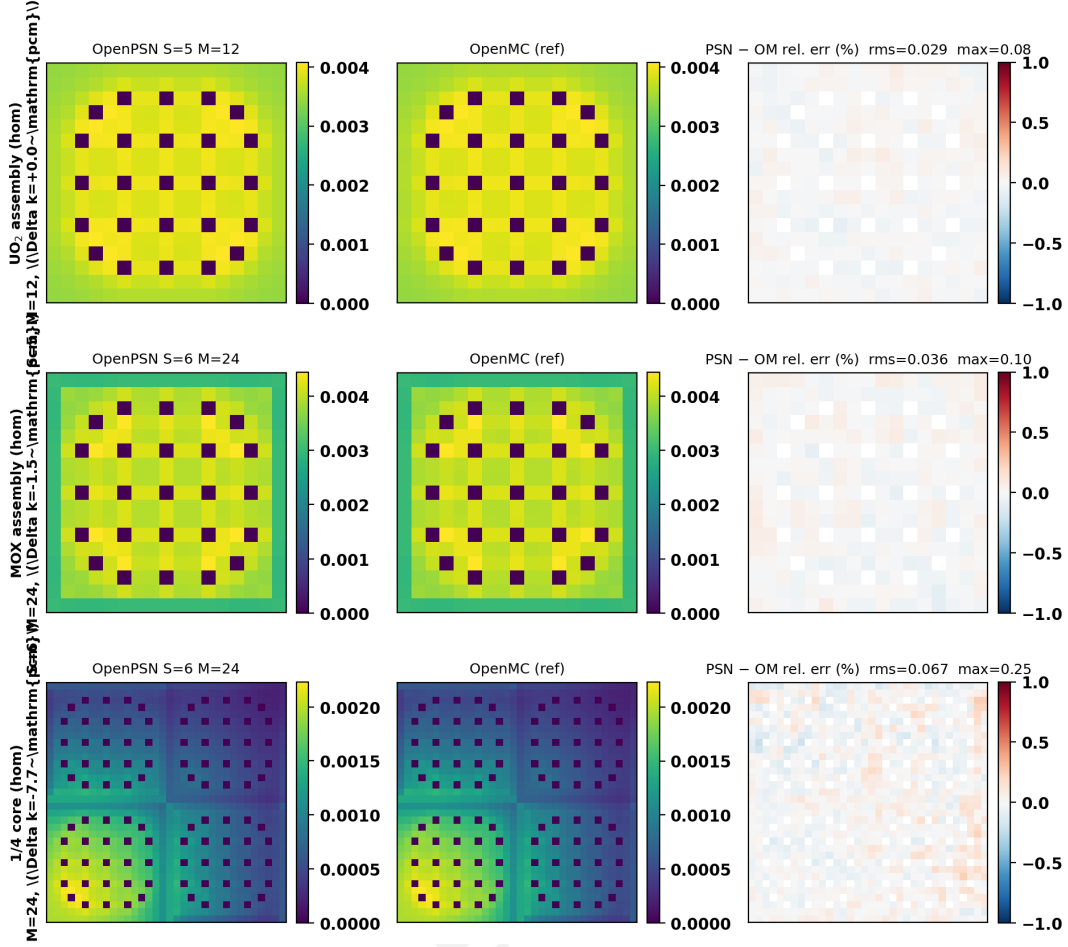


Figure 2: Pinwise-homogenised (BWW) best  $S \times M$  point per geometry — top:  $\text{UO}_2$  assembly,  $S=5, M=12, \Delta k = +0.0$  pcm; middle: MOX assembly,  $S=6, M=24, \Delta k = -1.5$  pcm; bottom: 1/4 core,  $S=6, M=24, \Delta k = -7.7$  pcm, cropped to the  $2 \times 2$  fuel block (the outer reflector-water region is omitted): pin-averaged fission source  $\nu\Sigma_f\phi$ , OpenPSN (left), OpenMC same-geometry reference (centre), pin-by-pin relative error on fuel pins (right; both maps fuel-normalised).

#### 4.4 Study II — the rectangular-node model with explicit intra-node structure

**Motivation and layout.** Study I isolates the discretisation error of the nodal PSN model, but its eigenvalue is still anchored to whatever homogenisation model feeds it. To remove that model from the problem we use the rectangular-node extension of Section 2.2: the circular C5G7 fuel rod ( $r = 0.54$  cm) is replaced by an equal-area fuel square of side  $s = \sqrt{\pi r^2} = 0.95713$  cm centred on the same 1.26 cm pitch, so that every pin cell becomes a  $3 \times 3$  array of rectangles (one fuel square plus eight water strips) and the whole core is a globally edge-aligned rectangular tiling. Figure 3 shows the layout. The pin pitch, the rod positions (including Gd pins, which are the centre rectangle of their  $3 \times 3$  block) and the total fuel area are unchanged; the layout was verified node by node — all 2601 fuel-centre rectangles of the 1/4 core coincide with the 2601 fuel pins of the homogenised grid, and the total area reproduces  $4129.35 \text{ cm}^2$  exactly. Un-homogenised 7-group cross sections (fuel, water, Gd, MOX) are then assigned directly to the rectangles. Two discretisation knobs are now open: the angular order  $M$  and the uniform spatial subdivision  $S$  (Section 2.2), and we sweep both.

Rectangularisation of the C5G7 fuel rod: circle  $\rightarrow$  equal-area square (pin pitch 1.26 cm and rod positions unchanged; only the centre rectangle of each rod carries the fuel material)

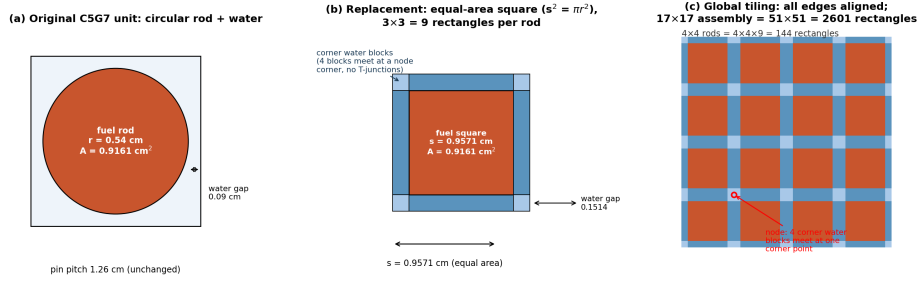


Figure 3: Rectangularisation of the C5G7 pin cell. (a) Original circular rod in its pitch cell (water gap 0.09 cm). (b) Equal-area fuel square ( $s = 0.95713$  cm,  $s^2 = \pi r^2$ ) with the pin cell split into  $3 \times 3$  rectangles; the four corner water blocks of neighbouring pins meet at a single node corner, so the tiling has no T-junctions. (c) Global tiling: a  $17 \times 17$  assembly is  $51 \times 51 = 2601$  rectangles and the  $1/4$  core is  $153 \times 153 = 23409$ .

**$S \times M$  convergence of the  $3 \times 3$  tiling.** Table 3 and Figure 4 give the full  $S \times M$  sweep (UO<sub>2</sub>/MOX assemblies:  $S=1 \dots 6$ ;  $1/4$  core:  $S=1, 2$ , the  $S=3$  core grid being  $9 \times$  larger in node count and only covered to  $M=8$ ; un-homogenised material XS, every point converged to  $10^{-10}$  in  $|\Delta \kappa_{\text{eff}}|$ ) against the directly computed OpenMC run on the *identical* rectangular-rod geometry (Section 4.4, 100k particles  $\times$  20200 batches, collision estimator, fixed seed): the single UO<sub>2</sub> assembly ( $51 \times 51$  rect. nodes, all edges reflective, ref.  $k = 1.3347260$ ), the MOX assembly (same size, ref.  $k = 1.1851053$ ), and the  $1/4$  core ( $153 \times 153$  rect. nodes, reflective on the symmetry edges, vacuum on the open edges, ref.  $k = 1.1872176$ ).

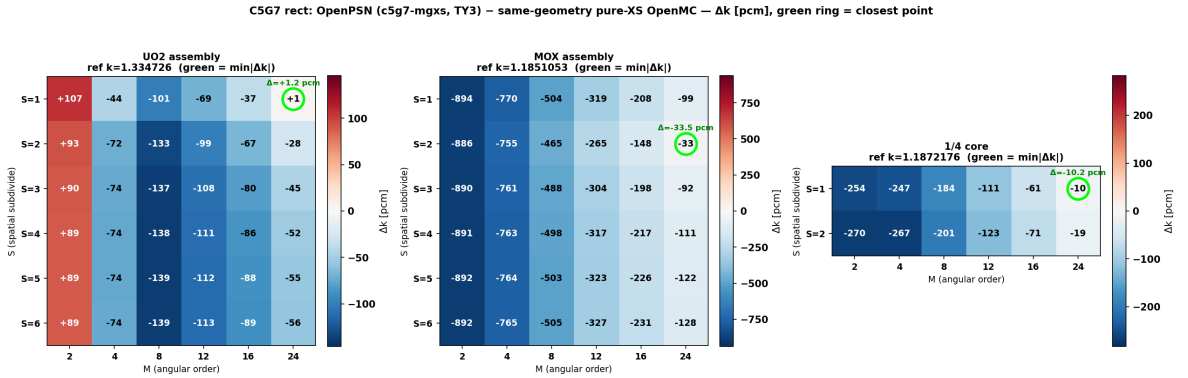


Figure 4: Same data as Table 3,  $\Delta k$  in pcm versus the same-geometry pure-XS OpenMC reference; colour bars symmetric about zero, green ring = point of smallest  $|\Delta k|$  (UO<sub>2</sub>:  $S=1, M=24$ , +1.2 pcm; MOX:  $S=2, M=24$ , -33.5 pcm; core:  $S=1, M=24$ , -10.2 pcm).

Figure 5 shows the fission field at the best available configuration per geometry (Table 3): the UO<sub>2</sub> and core rows agree with OpenMC at the sub-percent level, while the MOX row — at its best subdivision  $S=2$ , still -33.5 pcm off in  $\kappa_{\text{eff}}$  — carries that eigenvalue bias as a spatially structured field-level deviation.

Table 3: Rectangular-node OpenPSN,  $3 \times 3$  pin tiling:  $\text{UO}_2/\text{MOX}$  assemblies at  $S=1 \dots 6$ ,  $1/4$  core at  $S=1, 2$  (at  $S=3$  the core grid grows  $9 \times$ , only  $M=2, 4, 8$  were computed).  $\Delta k$  in pcm versus the same-geometry pure-XS OpenMC reference ( $\text{UO}_2$  assembly 1.3347260, MOX assembly 1.1851053,  $1/4$  core 1.1872176). Green marker in Figure 4 = smallest  $|\Delta k|$ .

|                                | $M=2$  | $M=4$  | $M=8$  | $M=12$ | $M=16$ | $M=24$ |
|--------------------------------|--------|--------|--------|--------|--------|--------|
| <i>UO<sub>2</sub> assembly</i> |        |        |        |        |        |        |
| $S=1$                          | +106.6 | -43.9  | -101.0 | -69.3  | -37.2  | +1.2   |
| $S=2$                          | +93.1  | -71.7  | -132.6 | -99.3  | -66.9  | -28.1  |
| $S=3$                          | +90.2  | -74.0  | -136.9 | -108.4 | -80.2  | -45.2  |
| $S=4$                          | +89.2  | -74.3  | -138.4 | -111.2 | -85.8  | -51.9  |
| $S=5$                          | +89.0  | -74.1  | -138.8 | -111.9 | -88.0  | -55.0  |
| $S=6$                          | +88.9  | -74.1  | -139.0 | -112.6 | -89.0  | -56.5  |
| <i>MOX assembly</i>            |        |        |        |        |        |        |
| $S=1$                          | -893.6 | -769.8 | -503.5 | -319.0 | -207.9 | -98.9  |
| $S=2$                          | -885.5 | -754.9 | -464.9 | -265.5 | -148.3 | -33.5  |
| $S=3$                          | -889.5 | -760.7 | -488.3 | -304.3 | -198.3 | -92.2  |
| $S=4$                          | -891.2 | -762.8 | -498.0 | -316.9 | -216.8 | -111.2 |
| $S=5$                          | -891.9 | -763.9 | -502.6 | -322.9 | -226.0 | -122.4 |
| $S=6$                          | -892.3 | -764.6 | -505.1 | -327.4 | -231.0 | -128.4 |
| <i>1/4 core</i>                |        |        |        |        |        |        |
| $S=1$                          | -254.3 | -246.6 | -184.3 | -110.9 | -61.1  | -10.2  |
| $S=2$                          | -269.5 | -266.7 | -201.0 | -122.7 | -71.3  | -18.8  |

$3 \times 3$  rectangular tiling, best  $S \times M$  configuration per geometry (Table 3) — pin-averaged fission  $\Sigma_f \phi$  vs. same-geometry OpenMC

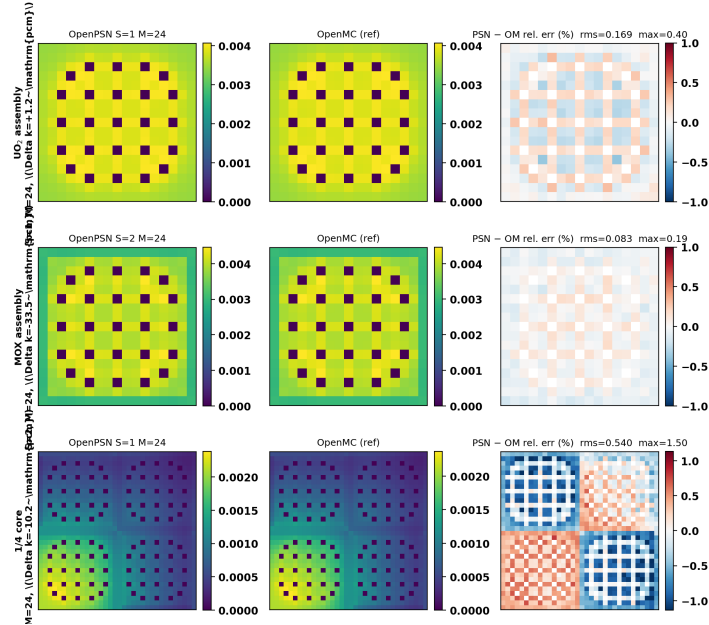


Figure 5:  $3 \times 3$  rectangular tiling, best  $S \times M$  configuration per geometry (Table 3): pin-averaged fission  $\Sigma_f \phi$ , OpenPSN (left), OpenMC on the identical rectangular-rod geometry (centre), pin-by-pin relative error on fuel pins (right; both maps normalised to unit sum). Top:  $\text{UO}_2$  assembly,  $S=1, M=24$ ,  $\Delta k = +1.2$  pcm; middle: MOX assembly,  $S=2, M=24$ ,  $\Delta k = -33.5$  pcm; bottom:  $1/4$  core,  $S=1, M=24$ ,  $\Delta k = -10.2$  pcm (fuel region).

The sweep shows three things. (i) On the  $\text{UO}_2$  assembly the best point ( $S=1, M=24, +1.2$  pcm) lies within the OpenMC standard error ( $\pm 3.6$  pcm): with a low-absorption fuel the nodal model is essentially reference-grade on this geometry, and every deeper subdivision is worse, monotonically from  $-28.1$  pcm at  $S=2$  to  $-56.5$  pcm at  $S=6$  — spatial subdivision is neutral-to-harmful. (ii) On the MOX assembly the tiling as constructed ( $S=1$ ) is badly off even at  $M=24$  ( $-98.9$  pcm); a single subdivision to  $S=2$  improves it to its best point ( $-33.5$  pcm), and any further subdivision ( $S=3\dots 6$ ) degrades it again, back to  $-128.4$  pcm — the non-monotonic  $S$ -response with a single interior optimum at  $S=2$  is the clearest signature of the gap-node problem below. (iii) On all three geometries the  $M$ -convergence is non-monotonic at low order (an  $M=8$  dip of  $-101\dots -504$  pcm) and only flattens out at  $M=24$ ; on the core the best point is  $S=1, M=24$  ( $-10.2$  pcm), again with  $S=2$  slightly worse.

**The slender water gap.** The  $S$ -saturation pattern above points to a geometric cause. In the  $3 \times 3$  tiling the water-gap nodes are intrinsically slender: each edge strip node measures  $s/3 \times w = 0.3190 \times 0.1514$  cm, an aspect ratio of 2.1:1 (the corner nodes,  $w \times w$ , are square but carry little physics). The uniform  $S$  subdivision divides *both* dimensions by  $S$ , so every sub-node inherits the parent’s aspect ratio — increasing  $S$  refines the slender nodes but never makes them less slender. The residual eigenvalue bias of Table 3 is therefore plausibly anchored in the strip nodes, in a part of the mesh that  $S$ -subdivision structurally cannot repair; this is consistent with the  $\text{UO}_2$  rows that degrade smoothly and monotonically as  $S$  grows at fixed  $M$  (weak water coupling: subdivision adds nodes but never re-shapes the strips), the MOX rows whose only relief is a single subdivision — best  $S=2$ , degrading again for  $S \geq 3$  (strong fuel–water coupling in the gap) — and the  $1/4$  core, whose  $S=2$  row is uniformly worse than  $S=1$  across all  $M$ . The remedy is a tiling in which the water-gap nodes are themselves close to square — obtained by a finer, quasi-square fuel split rather than by  $S$ . We first isolate this effect on a single pin.

**Single-pin validation on the quasi-square ( $6 \times 6$ ) split.** As a controlled experiment we take the single  $\text{UO}_2$  pin cell of the same rectangularised layout and split the equal-area fuel square  $6 \times 6$  instead of  $3 \times 3$ , giving an  $8 \times 8$  base grid per pin cell ( $6 \times 6$  fuel sub-nodes of side  $s/6 = 0.15952$  cm plus water strips of  $0.15952 \times 0.15144$  cm, aspect ratio 1.05:1 — every node of the cell is now close to square, **as illustrated in Figure 6**). The single cell with all-reflective boundaries is converged to  $10^{-10}$  in  $|\Delta\kappa_{\text{eff}}|$  on a  $S \times M$  sweep of 60 points ( $S = 1\dots 6, M = 2\dots 192$ ), against an independently computed OpenMC reference on the same  $8 \times 8$  geometry (4000 batches,  $k = 1.3267918 \pm 8.2$  pcm). Table 4 and Figure 7 summarise the sweep.

Single  $\text{UO}_2$  pin cell (1.26 cm pitch, all-reflective): tiling of the equal-area fuel square  $s^2 = \pi r^2$ .  $S$ -subdivision refines nodes without changing their shape; the manual  $6 \times 6$  split makes the water strips near-square.

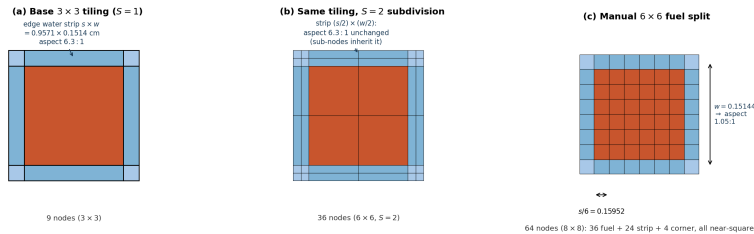


Figure 6: Single  $\text{UO}_2$  pin cell (1.26 cm pitch, all-reflective): tiling of the equal-area fuel square  $s^2 = \pi r^2$ . (a) Base  $3 \times 3$  tiling: edge water strips  $s \times w = 0.9571 \times 0.1514$  cm (aspect 6.3:1). (b) The same tiling with  $S=2$  subdivision: 36 nodes, strips  $(s/2) \times (w/2)$ , aspect unchanged —  $S$  refines but never reshapes the nodes. (c) Manual  $6 \times 6$  fuel split:  $8 \times 8$  base grid (64 nodes), strips  $s/6 \times w = 0.15952 \times 0.15144$  cm, aspect 1.05:1; every node is now close to square.

Table 4: Single  $\text{UO}_2$  pin cell,  $6 \times 6$  fuel split ( $8 \times 8$  base grid, all-reflective):  $\Delta k$  in pcm versus the OpenMC reference  $1.3267918 \pm 8.2$ . Green ring = best point  $M=192$ ,  $S=6$  ( $-8.7$  pcm, inside the reference’s own standard error).

|       | $M=2$ | $M=4$  | $M=8$  | $M=12$ | $M=16$ | $M=24$ | $M=48$ | $M=72$ | $M=96$ | $M=192$ |
|-------|-------|--------|--------|--------|--------|--------|--------|--------|--------|---------|
| $S=1$ | -73.0 | -182.9 | -208.2 | -165.1 | -132.0 | -88.0  | -41.6  | -28.2  | -22.6  | -16.5   |
| $S=2$ | -72.9 | -178.1 | -197.7 | -154.5 | -119.7 | -78.6  | -36.3  | -23.0  | -17.3  | -10.7   |
| $S=3$ | -72.6 | -176.8 | -195.1 | -151.9 | -116.7 | -75.9  | -35.1  | -21.8  | -16.2  | -9.6    |
| $S=4$ | -72.5 | -176.3 | -193.9 | -150.8 | -115.4 | -74.7  | -34.3  | -21.1  | -15.7  | -9.1    |
| $S=5$ | -72.4 | -176.0 | -193.4 | -150.2 | -114.7 | -74.1  | -33.9  | -20.7  | -15.4  | -8.9    |
| $S=6$ | -72.4 | -175.8 | -193.0 | -149.9 | -114.3 | -73.7  | -33.6  | -20.5  | -15.2  | -8.7    |

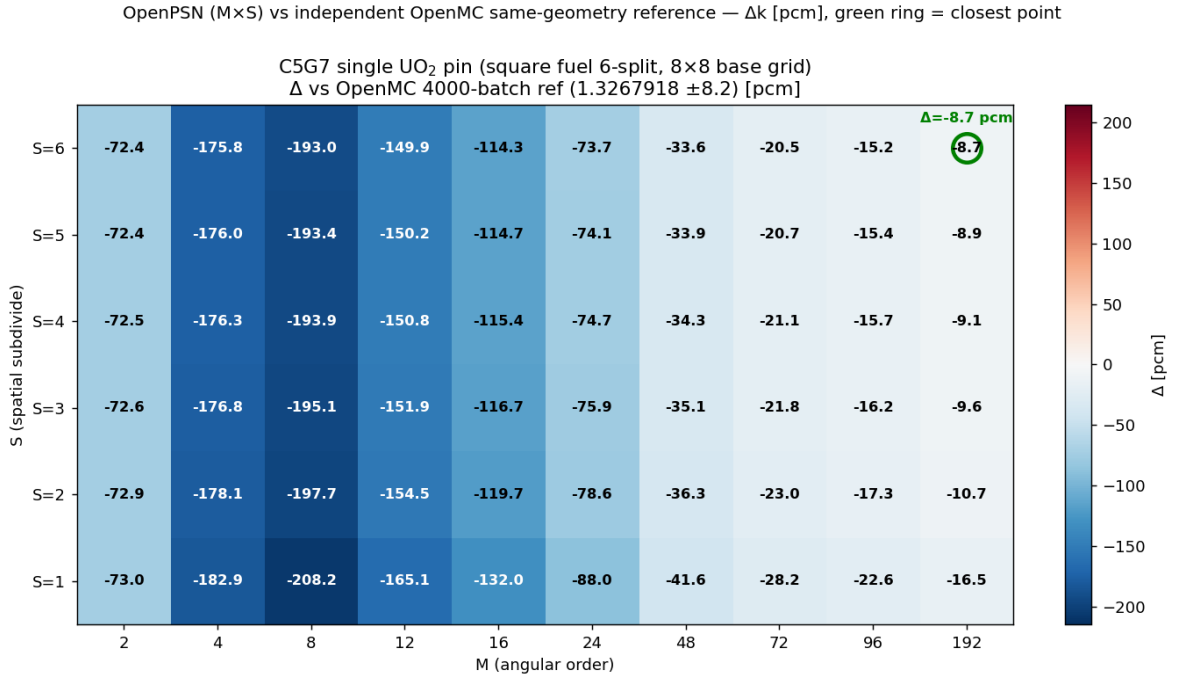


Figure 7: Single-pin  $6 \times 6$ -split  $S \times M$  sweep,  $\Delta k$  in pcm versus the OpenMC same-geometry reference; green ring = best point ( $M=192$ ,  $S=6$ ,  $\Delta = -8.7$  pcm).

On the quasi-square single pin the model converges to within the reference’s own Monte Carlo standard error ( $-8.7$  pcm at  $M=192$ ,  $S=6$ ) — and the spatial field agrees even more tightly: at the best configuration the node-level fission-rate map differs from OpenMC by 0.05% RMS (max 0.10%) and the scalar flux by 0.01% RMS (max 0.03%) over the fuel nodes (Figure 8). The convergence is smooth in  $M$  once the  $M=8$  angular damped-oscillation dip is passed, and  $S$  contributes a further  $\sim 8$  pcm of improvement from  $S=1$  to  $S=6$  that is still active at  $M=192$  — i.e. on a quasi-square mesh both knobs are effective. This contrasts with the  $3 \times 3$  tiling, where the  $S=1 \rightarrow 2$  change buys at most a factor of three on the MOX bias and the best subdivision is geometry-dependent (Table 3).

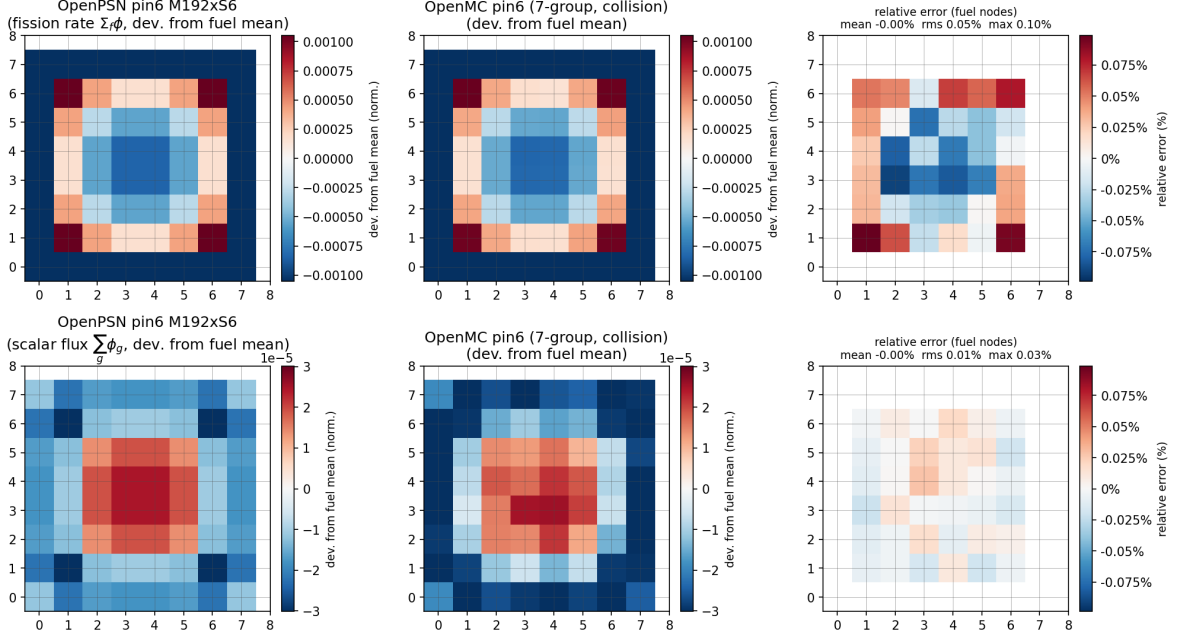


Figure 8: Single-pin 6  $\times$  6 split, best configuration ( $M=192$ ,  $S=6$ ): node-level ( $8 \times 8$ ) fission rate  $\Sigma_f \phi$  (top) and scalar flux  $\sum_g \phi_g$  (bottom), OpenPSN vs. OpenMC (7-group, collision estimator) with the pin-by-pin relative error on the fuel nodes; deviation from the fuel-node mean, maps normalised to unit fuel sum. Fission: mean  $-0.00\%$ , RMS  $0.05\%$ , max  $0.10\%$ ; flux: RMS  $0.01\%$ , max  $0.03\%$ .

**Quasi-square tiling of the full geometries.** We now apply the single-pin quasi-square split to the full benchmark: every fuel square of the  $3 \times 3$  tiling is subdivided  $6 \times 6$ , so each pin cell becomes an  $8 \times 8$  block of rectangles — fuel sub-nodes  $0.15952 \times 0.15952$  cm and water strips  $0.15952 \times 0.15144$  cm, aspect ratio 1.05:1. The  $17 \times 17$  assembly becomes  $136 \times 136 = 18496$  rectangles and the  $1/4$  core  $408 \times 408 = 166464$ ; the pin-centre material layout, the rod positions, the fuel area and the total area are unchanged (verified node by node, max cell-width deviation 0.0). Because the physical geometry is identical to the rectangular-rod problem of Section 4.4, the *existing* same-geometry OpenMC references (1.3347260, 1.1851053, 1.1872176) apply without re-running: the comparison is again a pure method difference.

**$M$ -convergence of the quasi-square tiling.** Table 5 and Figure 9 give the  $S=1$  sweep,  $M = 2 \dots 96$  on the two assemblies (core  $M \geq 24$  and the  $M=192$  extension in progress). Two features stand out. First,  $M$ -convergence is markedly faster than on the  $3 \times 3$  tiling at comparable resolution: on the MOX assembly the bias drops from  $-893.7$  to  $+2.1$  pcm over  $M=2 \dots 96$ , whereas the  $3 \times 3$  tiling reaches  $-98.9$  pcm at  $M=24$  ( $-33.5$  pcm only with  $S=2$ ); the low-order  $M=8$  dip remains but is shallower on the core. Second, the pin-averaged fission distribution converges *far* ahead of the eigenvalue: on both assemblies the pin-by-pin RMS error against OpenMC is at or below  $0.09\%$  from  $M=8$  onward ( $0.056\%$  UO<sub>2</sub> and  $0.069\%$  MOX at  $M=96$ ), while  $\Delta k$  still carries  $-161 \dots -28.7$  pcm of bias on UO<sub>2</sub> and  $-512 \dots +2.1$  pcm on MOX over the same range — the intra-fuel quasi-square split delivers the full spatial shape of the solution at a fraction of the angular order needed for the global eigenvalue.

Table 5: Quasi-square ( $6 \times 6$  fuel split) OpenPSN,  $S=1$ :  $\Delta k$  in pcm versus the same-geometry pure-XS OpenMC reference (refs. as in Table 3). Dombey two-point source extrapolation on MOX/core. Core  $M \geq 24$  and the  $M=192$  extension pending.

|                                | $M=2$  | $M=4$  | $M=8$  | $M=12$ | $M=16$ | $M=24$ | $M=48$ | $M=96$ |
|--------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|
| <i>UO<sub>2</sub> assembly</i> | +86.2  | −85.6  | −160.7 | −138.4 | −115.7 | −82.6  | −44.9  | −28.7  |
| <i>MOX assembly</i>            | −893.7 | −768.5 | −511.6 | −333.2 | −235.8 | −129.2 | −33.2  | +2.1   |
| <i>1/4 core</i>                | −278.8 | −281.5 | −232.4 | −166.9 | −126.2 | ...    | ...    | ...    |

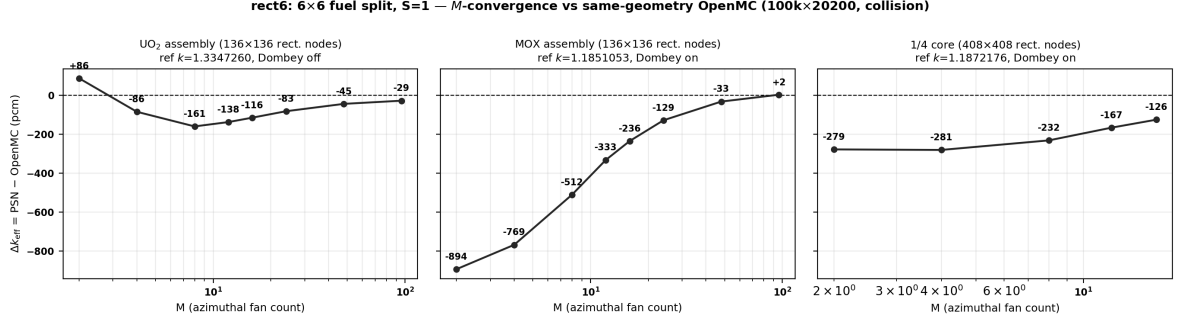


Figure 9: Quasi-square tiling,  $S=1$ :  $\Delta k$  versus  $M$  against the same-geometry OpenMC reference (100k×20200, collision estimator). UO<sub>2</sub> assembly (left) shows the non-monotonic low-order behaviour; the MOX assembly (centre) and 1/4 core (right) converge monotonically with  $M$ .

Figure 10 compares the pin-averaged fission maps  $\Sigma_f \phi$  of the best quasi-square configuration (smallest  $|\Delta k|$  per geometry, currently  $M=96$  on both assemblies) against the OpenMC reference on the identical geometry, using the same  $\nu$ -correction and fuel-block normalisation as Section 4.4. On the UO<sub>2</sub> assembly the pin-by-pin difference is a mean of +0.002% and an RMS of 0.056% (max 0.19%, p95 0.10%); on the MOX assembly a mean of −0.006% and an RMS of 0.069% (max 0.21%, p95 0.12%) — i.e. the intra-fuel quasi-square tiling matches the reference pin power to well under one tenths of a percent on both assemblies, with the residual confined to a few corner pins.

rect6 best configuration (min  $|\Delta k|$ ) — pin-averaged fission  $\Sigma_f \phi$  vs. same-geometry OpenMC

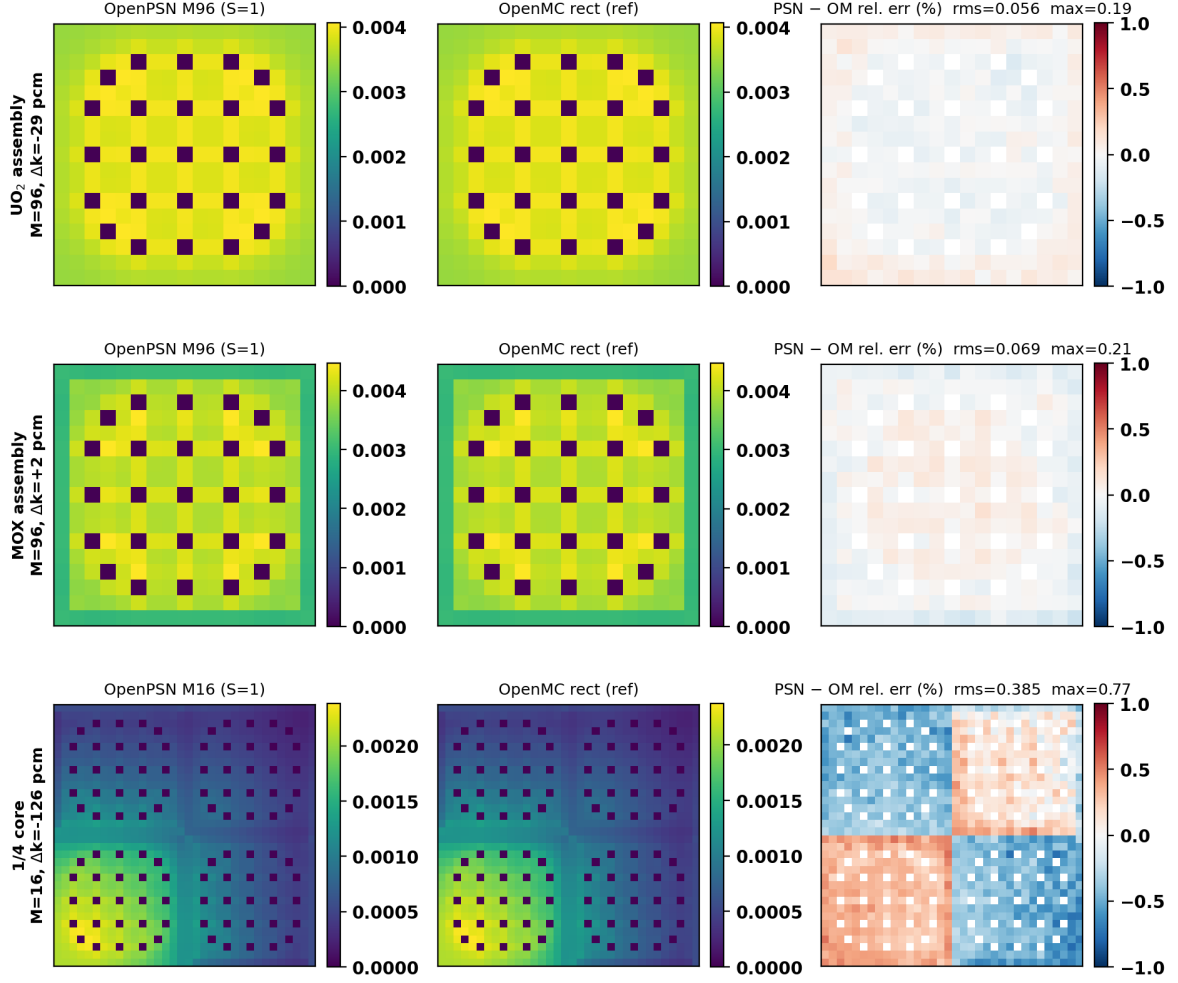


Figure 10: Quasi-square tiling, best configuration per geometry (min  $|\Delta k|$ ,  $S=1$ ): pin-averaged fission  $\Sigma_f \phi$ , OpenPSN (left), OpenMC same-geometry reference (centre), pin-by-pin relative error on fuel pins (right; both maps fuel-block normalised). Top:  $\text{UO}_2$  assembly,  $M=96$ ; middle: MOX assembly,  $M=96$ ; bottom: 1/4 core (interim best  $M=16$ ,  $M \geq 24$  in progress), cropped to the  $2 \times 2$  fuel block — the outer reflector-water region is omitted.

**Rectangular-rod reference.** The circular pin cell is the benchmark’s reference geometry, while the PSN nodes tile the same cell with an equal-area square fuel block (side 0.9571 cm, water gap 0.1514 cm). To separate the geometric effect from the method difference we run OpenMC on exactly that rectangular-rod geometry (same un-homogenised data, same collision-estimator pin tallies, 20200 batches  $\times 10^5$  particles, fixed seed): the square-rod  $\text{UO}_2$  assembly returns  $\kappa_{\text{eff}} = 1.3347260$  ( $\pm 3.6$  pcm), the MOX assembly 1.1851053 ( $\pm 3.3$  pcm, identical clone of the  $\text{UO}_2$  run with only the pin material grid changed to the C5G7 MOX layout) and the 1/4 core 1.1872176 ( $\pm 3.4$  pcm). Against these same-geometry references the  $3 \times 3$  tiling behaves as in Table 3 and Figure 5; the pin-level fission maps  $\Sigma_f \phi$  of the  $3 \times 3$  tiling at  $S=1$ ,  $M=24$  agree with OpenMC to 0.17% RMS (max 0.40%) on the  $\text{UO}_2$  assembly and 0.54% RMS (max 1.50%) over all 1056 fuel pins of the 1/4 core, while the MOX assembly carries the  $-98.9$  pcm eigenvalue bias as a spatially structured field-level deviation; the quasi-square tiling holds at or below 0.09% RMS on both assemblies from  $M \geq 8$  and settles at 0.06–0.07% by  $M=96$  (Figure 10).

#### 4.5 Comparison with the published diffusion / SN / nodal landscape

The original NEA report [2] tabulates 20 deterministic codes on C5G7-2D. Table 6 reproduces the diffusion- and nodal-family values we could extract from it, together with recent P1/SP3/SN/nodal literature results and our own OpenPSN result.

Table 6: C5G7-2D  $\kappa_{\text{eff}}$  by method class (2-D multi-group, homogenised pin model unless noted). NEA-2003  $k$  values (its Table 17) plus recent P1/SP3 literature.  $\Delta = (k - 1.18646) \times 10^5$  in pcm, except where stated (STELLA vs its PEACH reference 1.18716). The original NEA table prints the relative per-cent error against its MCNP reference 1.18655; the pcm values here are recomputed from the printed  $k$  against 1.18646.

| method class                                               | code          | $\kappa_{\text{eff}}$ | $\Delta$ (pcm)  |
|------------------------------------------------------------|---------------|-----------------------|-----------------|
| <i>Pure P1 diffusion</i>                                   |               |                       |                 |
| diffusion (FEM)                                            | CRONOS2       | 1.18323               | −323            |
| P1 (pin-hom)                                               | STELLA-P1 [9] | 1.18426               | −290 (vs PEACH) |
| <i>SN</i>                                                  |               |                       |                 |
| SN-FEM                                                     | CRONOS2-SN    | 1.18338               | −308            |
| SN-FDM                                                     | DORT-GRS      | 1.18482               | −164            |
| SN-FDM                                                     | DORT-ORNL     | 1.18496               | −150            |
| SN-FDM                                                     | TWODANT       | 1.18668               | +22             |
| SN (diamond diff.)                                         | PARTISN       | 1.18637               | −9              |
| SN (unstructured)                                          | PERICLES      | 1.18658               | +12             |
| <i>High-order nodal <math>P_n</math> / surf. harmonics</i> |               |                       |                 |
| $P_n$ -nodal (FEM)                                         | VARIANT-SE    | 1.18495               | −151            |
| $P_n$ -nodal + int. transp.                                | VARIANT-ISE   | 1.18745               | +99             |
| surface harm. (G3/P2)                                      | SUHAM-2D      | 1.18628               | −18             |
| <i>MOC</i>                                                 |               |                       |                 |
| MOC                                                        | CHAPLET       | 1.18656               | +10             |
| MOC                                                        | MCCG3D        | 1.18657               | +11             |
| MOC                                                        | DeCART        | 1.18660               | +14             |
| MOC                                                        | APOLLO2       | 1.18618               | −28             |
| MOC                                                        | CRX           | 1.18813               | +167            |
| MOC (stoch. rays)                                          | UNKGRO        | 1.18523               | −123            |
| <i>This work</i>                                           |               |                       |                 |
| PSN nodal (rect. node, $M=16$ )                            | OpenPSN       | 1.1866067             | +14.7           |

*Readings.*

- **Pure P1 diffusion is systematically low** — CRONOS2 −323 pcm (−0.27%), STELLA-P1 −290 pcm vs its PEACH MOC reference 1.18716: no transport correction and no self shielding.
- **SN spans −308 (CRONOS2-SN) to +22 pcm (TWODANT); DORT −164/−150; the better-differenced / unstructured variants PARTISN (−9) and PERICLES (+12) within  $\pm 0.01\%$ .**
- **High-order nodal  $P_n$  / surface harmonics** spans −151 (VARIANT-SE) to +99 pcm (VARIANT-ISE); SUHAM-2D −18; VARIANT-ISE also has the best pin power in the

original table (MRE 0.11%).

- **MOC (explicit pin geometry) clusters within  $\pm 28$  pcm** (CHAPLET +10, MCCG3D +11, DeCART +14, APOLLO2 -28), with two outliers (CRX +167, UNKGRO -123); TWODANT is SN-FDM (+22); the high-fidelity Rattlesnake converged limit is 1.186446 ( $\sim 1$  pcm from the 1.186456 reference).
- **OpenPSN in this work.** The two C5G7 studies of Sections 4.3–4.4 bound the answer from both sides. With the self-shielding-aware BWW pinwise homogenisation, the best points of the  $S \times M$  sweep are +0.0 pcm (UO<sub>2</sub> assembly), -1.5 pcm (MOX assembly) and -7.7 pcm (1/4 core) vs. the OpenMC homogenised references — the two assemblies inside their own Monte Carlo standard errors; with the rectangular-node model that removes homogenisation entirely,  $M=16$  lands at +11.1 pcm vs. the OpenMC explicit-geometry reference ( $\sim 3.3$  its  $\pm 3.4$  pcm standard error) and +14.7 pcm vs. 1.18646, and at -61.1 pcm vs. the OpenMC run on the identical rectangular-rod geometry (Section 4.4); the pin-level fission maps agree within 0.6% RMS on both geometries. In either case OpenPSN sits inside the MOC reference cluster ( $\pm 37$  pcm, plus the better-differenced SN and unstructured codes), and the rectangular-node result is within the  $M=16$  angular convergence of the nodal PSN method on this problem.

Two published outliers are worth noting for completeness: COHINT ( $P_2$  interface currents, -1125 pcm) and HELIOS (collision probability, +675 pcm), both far outside the converged cluster and attributable to their respective current/transport approximations.

## 5 Conclusions

We built **OpenPSN**, a clean, multi-group, sparse, vectorised implementation of the Phase Space Nodal method, and validated it on two levels. On the source paper’s problems it reproduces all 248 published eigenvalue data points to a few pcm (full point-by-point comparison in Appendix A) while removing the dense-formulation memory blow-up (exact to  $10^{-15}$ ; the 10404-node C5G7 case assembles in  $\sim 54$  MB and converges in  $\sim 54$  min on one core, where the dense form would need a  $\sim 504$  GB matrix). We also extended the node operators to arbitrary per-node widths, giving the first PSN implementation with explicit, un-homogenised intra-node fuel/water structure.

On the C5G7 MOX benchmark the two studies separate the two error sources that a nodal code carries. In the *pinwise-homogenised* study, the  $S \times M$  sweep against a directly computed OpenMC reference on the same homogenised data shows the discretisation converging cleanly: both assemblies reach the Monte Carlo standard error ( $\approx 3.3$ – $3.5$  pcm) by  $S=5$ – $6$ ,  $M \geq 8$  (+0.0 pcm UO<sub>2</sub>, -1.5 pcm MOX), and the best core point (with the self-shielding-aware BWW recipe) is -7.7 pcm versus the OpenMC homogenised reference, i.e. +28.6 pcm versus the high-fidelity reference 1.18646. That residual is the homogenisation model itself, and the best-point pin-power fields agree with the same-recipe OpenMC maps to 0.07% RMS (max 0.25%): a smooth method-level bias, not a cancellation of local errors. In the *rectangular-node* study the homogenisation model is removed entirely: with un-homogenised 7-group material data the 1/4-core eigenvalue converges monotonically with  $M$  against the OpenMC same-geometry reference ( $-184.3 \rightarrow -110.9 \rightarrow -61.1 \rightarrow -10.2$  pcm for  $M = 8 \rightarrow 24$ ,  $S=1$ ), the best point  $S=1$ ,  $M=24$  sits -10.2 pcm from that reference (itself +75.8 pcm above 1.18646, the pure geometric effect of the rectangular-rod idealization), and the pin-averaged fission map  $\Sigma_f \phi$  (the same quantity as the Monte Carlo fission tally) agrees with OpenMC to 0.17% RMS on the UO<sub>2</sub> assembly and 0.54% over all 1056 fuel pins of the core (max 1.50%; per-assembly maxima 0.71–1.50%, largest in the MOX cells where the local gradients are steepest). Placed against the 20-code NEA landscape and the recent P1/SP3/SN/nodal literature, both OpenPSN results sit inside

the high-fidelity MOC reference cluster: the nodal PSN method, once its homogenisation model is controlled (by a self-shielding-aware recipe or by explicit rectangular nodes), reproduces resolved-pin transport at a fraction of the mesh cost of discrete-ordinates or characteristics methods.

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## A Full point-by-point reproduction of the published PSN results

Table 1 summarised the maxima. Tables 7–10 give the complete 240-cell comparison for the three printed eigenvalue grids (checkerboard weak/strong, BWR without Gd) plus the with-Gd valley, in the same “this work / paper ( $\Delta$ ) cell format used in the reproduction notes shipped in `snapshot/`. Row labels:  $S$  = spatial subdivision of the checkerboard grid,  $N$  = spatial grid of the BWR bundle. The remaining 8 points (restricted  $M = 24$ ,  $S = 1, 2, 4, 8$ , weak/strong) agree with the printed Figs. 4–5 within the plot-reading uncertainty and are listed in `snapshot/data/report_data.json`.

Table 7: Fig. 3-A, weak absorption, generic ( $I = 30$ ):  $\Delta$  vs the printed values (reference  $\kappa_{\text{eff}} = 1.12974$ ).  $S$  = spatial subdivision.

| $S$            | $M=2$                | $M=4$                | $M=8$               | $M=12$             | $M=16$             | $M=24$            |
|----------------|----------------------|----------------------|---------------------|--------------------|--------------------|-------------------|
| $1 \times 1$   | +136.2/136 (+0.2)    | +389.0/388 (+1.0)    | +703.7/703 (+0.7)   | +793.9/793 (+0.9)  | +835.2/835 (+0.2)  | +873.5/873 (+0.5) |
| $2 \times 2$   | −275.4/ − 276 (+0.6) | −143.6/ − 144 (+0.4) | +154.9/154 (+0.9)   | +217.2/217 (+0.2)  | +244.4/244 (+0.4)  | +268.2/268 (+0.2) |
| $3 \times 3$   | −374.4/ − 375 (+0.6) | −263.2/ − 264 (+0.8) | +19.1/18 (+1.1)     | +79.3/79 (+0.3)    | +104.4/104 (+0.4)  | +124.1/123 (+1.1) |
| $4 \times 4$   | −412.0/ − 413 (+1.0) | −309.4/ − 310 (+0.6) | −34.6/ − 35 (+0.4)  | +24.9/24 (+0.9)    | +48.2/47 (+1.2)    | +66.9/66 (+0.9)   |
| $5 \times 5$   | −430.1/ − 431 (+0.9) | −332.1/ − 333 (+0.9) | −61.2/ − 62 (+0.8)  | −2.7/ − 3 (+0.3)   | +20.2/19 (+1.2)    | +38.2/38 (+0.2)   |
| $6 \times 6$   | −440.3/ − 441 (+0.7) | −344.8/ − 346 (+1.2) | −76.3/ − 77 (+0.7)  | −18.5/ − 19 (+0.5) | +4.1/3 (+1.1)      | +21.8/21 (+0.8)   |
| $7 \times 7$   | −446.5/ − 447 (+0.5) | −352.7/ − 353 (+0.3) | −85.7/ − 86 (+0.3)  | −28.4/ − 29 (+0.6) | −6.1/ − 7 (+0.9)   | +11.4/11 (+0.4)   |
| $8 \times 8$   | −450.5/ − 451 (+0.5) | −357.9/ − 359 (+1.1) | −92.0/ − 93 (+1.0)  | −35.0/ − 36 (+1.0) | −12.9/ − 14 (+1.1) | +4.4/4 (+0.4)     |
| $9 \times 9$   | −453.4/ − 454 (+0.6) | −361.5/ − 362 (+0.5) | −96.3/ − 97 (+0.7)  | −39.6/ − 40 (+0.4) | −17.7/ − 18 (+0.3) | −0.5/ − 1 (+0.5)  |
| $10 \times 10$ | −455.4/ − 456 (+0.6) | −364.1/ − 365 (+0.9) | −99.5/ − 100 (+0.5) | −43.0/ − 44 (+1.0) | −21.2/ − 22 (+0.8) | −4.1/ − 5 (+0.9)  |

*Note:* each cell is “this work / paper ( $\Delta$ ),  $\Delta$  = this work − paper, in pcm; paper values as printed in Ref. [1]; the Fig. 10/12 references are read from the printed contour plots ( $\pm 1$ –2 pcm reading uncertainty).

Table 8: Fig. 3-B, strong absorption, generic ( $I = 30$ ):  $\Delta$  vs the printed values (reference  $\kappa_{\text{eff}} = 0.51673$ ).  $S$  = spatial subdivision.

| $S$            | $M=2$                  | $M=4$                  | $M=8$                | $M=12$               | $M=16$               | $M=24$              |
|----------------|------------------------|------------------------|----------------------|----------------------|----------------------|---------------------|
| $1 \times 1$   | +591.0/591 (+0.0)      | +1762.4/1763 (-0.6)    | +3199.2/3199 (+0.2)  | +3596.7/3597 (-0.3)  | +3776.6/3777 (-0.4)  | +3941.3/3941 (+0.3) |
| $2 \times 2$   | -1534.9/ - 1535 (+0.1) | -790.2/ - 790 (-0.2)   | +807.9/808 (-0.1)    | +1132.7/1133 (-0.3)  | +1271.3/1271 (+0.3)  | +1392.4/1393 (-0.6) |
| $3 \times 3$   | -2068.0/ - 2068 (+0.0) | -1425.6/ - 1425 (-0.6) | +113.2/113 (+0.2)    | +431.2/431 (+0.2)    | +563.2/563 (+0.2)    | +666.4/666 (+0.4)   |
| $4 \times 4$   | -2272.6/ - 2272 (-0.6) | -1676.0/ - 1676 (+0.0) | -170.0/ - 170 (+0.0) | +145.3/145 (+0.3)    | +269.2/269 (+0.2)    | +367.8/368 (-0.2)   |
| $5 \times 5$   | -2371.6/ - 2372 (+0.4) | -1799.5/ - 1799 (-0.5) | -311.9/ - 312 (+0.1) | -1.4/ - 1 (-0.4)     | +120.5/121 (-0.5)    | +216.1/216 (+0.1)   |
| $6 \times 6$   | -2427.0/ - 2427 (+0.0) | -1869.2/ - 1869 (-0.2) | -393.2/ - 393 (-0.2) | -86.1/ - 86 (-0.1)   | +34.1/34 (+0.1)      | +128.3/128 (+0.3)   |
| $7 \times 7$   | -2461.0/ - 2461 (+0.0) | -1912.3/ - 1912 (-0.3) | -444.1/ - 444 (-0.1) | -139.3/ - 139 (-0.3) | -20.7/ - 21 (+0.3)   | +72.6/72 (+0.6)     |
| $8 \times 8$   | -2483.3/ - 2483 (-0.3) | -1940.8/ - 1941 (+0.2) | -478.0/ - 478 (+0.0) | -175.1/ - 175 (-0.1) | -57.6/ - 58 (+0.4)   | +35.0/34 (+1.0)     |
| $9 \times 9$   | -2498.8/ - 2499 (+0.2) | -1960.6/ - 1961 (+0.4) | -501.8/ - 502 (+0.2) | -200.2/ - 200 (-0.2) | -83.6/ - 84 (+0.4)   | +8.4/7 (+1.4)       |
| $10 \times 10$ | -2510.0/ - 2510 (-0.0) | -1974.9/ - 1975 (+0.1) | -519.1/ - 519 (-0.1) | -218.6/ - 219 (+0.4) | -102.6/ - 103 (+0.4) | -11.1/ - 12 (+0.9)  |

Note: each cell is “this work / paper ( $\Delta$ ),  $\Delta$  = this work - paper, in pcm; paper values as printed in Ref. [1]; the Fig. 10/12 references are read from the printed contour plots ( $\pm 1$ -2 pcm reading uncertainty).

Table 9: Fig. 10, BWR 12-pin bundle without Gd (2-group):  $\Delta$  vs the values read from the printed contour plot (reference  $\kappa_{\text{eff}} = 1.18797$ ).  $N$  = spatial grid.

| $N$ | $M=4$             | $M=8$              | $M=12$               | $M=16$               | $M=20$               | $M=24$               |
|-----|-------------------|--------------------|----------------------|----------------------|----------------------|----------------------|
| 1   | +126.4/123 (+3.4) | -74.8/ - 78 (+3.2) | -122.2/ - 125 (+2.8) | -140.3/ - 144 (+3.7) | -149.1/ - 152 (+2.9) | -154.2/ - 157 (+2.8) |
| 2   | +206.5/203 (+3.5) | +28.3/25 (+3.3)    | -15.6/ - 19 (+3.4)   | -31.9/ - 35 (+3.1)   | -40.1/ - 43 (+2.9)   | -44.9/ - 48 (+3.1)   |
| 3   | +226.2/223 (+3.2) | +54.8/51 (+3.8)    | +12.5/9 (+3.5)       | -3.6/ - 7 (+3.4)     | -11.6/ - 15 (+3.4)   | -16.3/ - 20 (+3.7)   |
| 4   | +233.7/230 (+3.7) | +65.2/62 (+3.2)    | +23.8/20 (+3.8)      | +7.8/5 (+2.8)        | -0.1/ - 3 (+2.9)     | -4.6/ - 8 (+3.4)     |
| 5   | +237.4/234 (+3.4) | +70.4/67 (+3.4)    | +29.4/26 (+3.4)      | +13.6/10 (+3.6)      | +5.8/2 (+3.8)        | +1.3/ - 2 (+3.3)     |
| 6   | +239.4/236 (+3.4) | +73.3/70 (+3.3)    | +32.7/29 (+3.7)      | +16.9/14 (+2.9)      | +9.1/6 (+3.1)        | +4.7/1 (+3.7)        |
| 7   | +240.6/237 (+3.6) | +75.1/72 (+3.1)    | +34.7/31 (+3.7)      | +19.0/16 (+3.0)      | +11.3/8 (+3.3)       | +6.9/4 (+2.9)        |
| 8   | +241.4/238 (+3.4) | +76.3/73 (+3.3)    | +36.0/33 (+3.0)      | +20.4/17 (+3.4)      | +12.7/9 (+3.7)       | +8.3/5 (+3.3)        |
| 9   | +242.0/239 (+3.0) | +77.2/74 (+3.2)    | +37.0/34 (+3.0)      | +21.4/18 (+3.4)      | +13.7/10 (+3.7)      | +9.3/6 (+3.3)        |
| 10  | +242.4/239 (+3.4) | +77.8/74 (+3.8)    | +37.7/34 (+3.7)      | +22.1/19 (+3.1)      | +14.5/11 (+3.5)      | +10.1/7 (+3.1)       |

Note: each cell is “this work / paper ( $\Delta$ ),  $\Delta$  = this work - paper, in pcm; paper values as printed in Ref. [1]; the Fig. 10/12 references are read from the printed contour plots ( $\pm 1$ -2 pcm reading uncertainty).

Table 10: Fig. 12, BWR 12-pin bundle with Gd (2-group): this work only (the printed figure is a contour plot).  $N$  = spatial grid. Bold = best  $M$  in the row (min  $|\Delta|$ ); the valley shifts to lower  $M$  as  $N$  grows, exactly the Sec. 4.4 effect of the source paper. Best point overall:  $N = 9$ ,  $M = 20$ , +1.1 pcm.

| $N$ | $M=4$ | $M=8$       | $M=12$     | $M=16$     | $M=20$     | $M=24$ |
|-----|-------|-------------|------------|------------|------------|--------|
| 1   | -605  | <b>+311</b> | +565       | +681       | +746       | +788   |
| 2   | -998  | -152        | <b>+67</b> | +157       | +203       | +230   |
| 3   | -1140 | -253        | <b>-22</b> | +73        | +123       | +152   |
| 4   | -1207 | -283        | <b>-45</b> | +53        | +103       | +133   |
| 5   | -1264 | -318        | -72        | <b>+28</b> | +79        | +110   |
| 6   | -1310 | -351        | -98        | <b>+4</b>  | +57        | +88    |
| 7   | -1346 | -380        | -122       | <b>-18</b> | +36        | +67    |
| 8   | -1375 | -405        | -144       | -37        | <b>+17</b> | +49    |
| 9   | -1397 | -427        | -162       | -54        | <b>+1</b>  | +33    |
| 10  | -1415 | -445        | -178       | -69        | <b>-13</b> | +20    |

*Note:* each cell is “this work / paper ( $\Delta$ ),  $\Delta$  = this work – paper, in pcm; paper values as printed in Ref. [1]; the Fig. 10/12 references are read from the printed contour plots ( $\pm 1$ –2 pcm reading uncertainty).